

BRONTE 11 NNE, TX, USA

WMO: 747370

Lat: 32.0408N

Lon: 100.2495W

Elev: 1997

StdP: 13.67

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 03072

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	22.4	5.7	7.9	28.7	10.2	10.0	36.1	13.9	54.6	12.5	56.1	2.7	N/A	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	101.3	68.9	99.0	69.2	96.8	69.2	74.8	89.0	73.7	88.6	72.9	87.8	6.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	124.8	78.8	69.9	118.7	78.3	68.8	114.0	78.1	39.6	88.5	38.6	88.7	37.7	87.8	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.8	9.7	10.7	104.6	3.9	2.9	7.8	106.7	5.5	108.4	3.3	110.1	0.5	112.2		
(5)			9.6	75.6	4.1	1.4	6.6	76.6	4.2	77.4	2.0	78.2	-1.0	79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	43.3	48.7	57.2	64.7	71.9	80.3	81.9	82.0	75.5	64.3	53.6	45.2		
	DBStd	15.85	8.94	10.83	9.51	8.38	7.63	4.84	3.87	3.98	5.70	8.66	9.09	9.11		
	HDD50	714	242	141	44	7	0	0	0	0	0	10	67	203		
	HDD65	2619	671	461	269	105	27	0	0	0	4	119	350	613		
	CDD50	5871	36	104	267	447	678	909	989	994	764	452	176	55		
	CDD65	2302	0	4	27	95	240	459	524	529	318	96	9	1		
	CDH74	28393	35	157	483	1294	2755	5553	6702	6689	3341	1168	178	38		
	CDH80	14686	3	38	132	504	1298	3003	3817	3874	1594	390	30	3		
(14) Wind	WSAvg	4.3	3.9	4.6	5.1	5.6	5.1	5.0	4.0	3.6	3.5	3.8	3.6	3.7		
(15) Precipitation	PrecAvg	23.4	0.9	1.3	1.6	1.9	3.1	3.0	1.7	2.7	2.4	2.2	1.5	0.9		
	PrecMax	34.1	2.7	4.8	4.8	3.9	7.4	7.4	4.6	7.7	6.7	5.7	4.7	5.7		
	PrecMin	12.2	0.0	0.0	0.0	0.2	1.4	0.2	0.1	0.0	0.1	0.1	0.0	0.0		
	PrecStd	6.3	0.7	1.3	1.1	1.3	1.7	2.2	1.3	2.3	1.9	1.6	1.3	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.5	86.1	88.3	95.3	100.6	104.1	103.3	103.4	99.0	91.6	84.2	79.1		
		MCWB	54.6	57.4	59.4	60.4	65.9	68.8	68.6	69.1	67.8	66.8	60.6	56.2		
	2%	DB	72.7	78.8	83.8	90.4	96.2	100.3	100.0	101.0	95.3	88.1	78.9	72.5		
		MCWB	52.0	56.4	57.9	61.1	66.4	69.1	69.1	69.7	68.3	65.8	59.7	53.9		
	5%	DB	67.1	73.7	79.6	86.0	91.3	96.9	97.8	98.6	92.5	84.5	75.1	67.7		
		MCWB	49.8	54.8	57.9	61.4	66.2	69.4	69.7	70.1	68.3	64.5	58.4	51.9		
	10%	DB	61.7	68.5	75.3	81.7	87.3	93.9	95.5	96.0	89.2	81.0	70.9	62.7		
		MCWB	47.6	52.8	57.6	61.0	65.9	69.9	69.6	69.9	67.9	63.0	56.9	50.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	63.1	65.7	68.8	73.3	76.1	76.2	75.5	74.8	71.3	66.4	61.8		
		MCDB	69.3	73.6	78.4	82.5	87.4	90.1	89.5	90.7	85.9	83.9	77.5	70.1		
	2%	WB	54.5	60.5	63.9	67.3	71.9	74.4	74.6	74.2	72.9	69.8	63.9	58.1		
		MCDB	65.5	71.2	75.8	80.0	85.1	89.0	88.1	91.1	84.5	81.6	72.3	65.9		
	5%	WB	51.7	58.6	62.0	65.9	70.7	73.3	73.5	73.1	71.7	68.2	61.3	55.4		
		MCDB	64.2	69.2	72.4	78.2	83.2	87.9	87.7	90.8	83.5	79.0	69.7	64.1		
	10%	WB	48.8	55.1	59.8	64.2	69.3	72.2	72.5	72.0	70.8	66.4	59.2	51.8		
		MCDB	61.5	65.8	70.2	76.0	81.2	86.9	87.2	89.3	82.4	76.2	67.9	62.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	27.8	28.3	28.0	27.3	24.3	23.7	24.7	25.0	23.4	27.0	27.7	26.4		
		MCDBR	38.7	36.9	34.7	34.1	31.1	28.9	28.7	29.6	28.6	31.5	32.5	35.6		
		MCWBR	20.0	18.4	16.5	14.6	10.9	8.2	8.6	8.3	9.0	14.1	16.6	19.5		
		MCDBR	32.0	30.8	26.3	25.3	23.9	24.7	23.3	24.9	22.3	23.1	23.8	26.9		
	5% WB	MCWBR	19.0	18.7	14.1	13.0	10.0	8.6	7.7	7.5	7.3	11.2	14.5	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.300	0.318	0.363	0.382	0.400	0.413	0.398	0.372	0.349	0.316	0.296		
	taud		2.556	2.516	2.483	2.363	2.381	2.364	2.386	2.418	2.418	2.473	2.492	2.528		
	Ebn at Noon		301	306	306	294	287	280	276	280	284	284	285	290		
	Edh at Noon		26	30	33	39	39	39	38	36	35	31	28	25		
(44) All-Sky Solar Radiation	RadAvg		1036	1261	1638	1993	2085	2274	2209	2031	1694	1368	1088	918		
	RadStd		109	163	118	147	192	158	146	128	123	194	128	103		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+180

Nomenclature: See separate page